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414. ON POLYZOMAL CURVES.

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[Continued from page 522.]

129. In the case of a bicircular quartic, the two tangents at I and the two tangents at J meet in four points, which (although not recognising them as foci) I call the nodo-foci; these lie in pairs on two lines, diagonals of the quadrilateral formed by the four tangents (the third diagonal is of course the line IJ), which diagonals I call the "nodal axes;" and the point of intersection of the two nodal axes is the "centre" of the curve. The nodo-foci are four points, two of them real, the other two imaginary, viz., they are two pairs of antipoints, the lines through the two pairs respectively being, of course, the nodal axes; these are consequently real lines bisecting each other at right angles in the centre (with the relation $1 : i$ between the distances). The centre may also be defined as the intersection of the harmonic of IJ in regard to the tangents at I, and the harmonic of this same line in regard to the tangents at J. Speaking of the tangents as asymptotes, the nodo-foci are the angles of the rhombus formed by the two pairs of parallel asymptotes; the nodal axes are the diagonals of this rhombus, and the centre is the point of intersection of the two diagonals; as such it is also the intersection of the two lines drawn parallel to and midway between the lines forming each pair of parallel asymptotes.

Article No. 130. Circular Cubic and Bicircular Quartic; the Nodal or Symmetrical Case.

130. In a circular cubic or bicircular quartic, the pencil of the tangents from I and that of the tangents through J, considered as corresponding to each other in some one of the four arrangements, may be such that the line IJ considered as belonging to the two pencils respectively shall correspond to itself, and when this is so, the four foci, A, B, C, D, which are the intersections of the corresponding tangents in question, will lie in a line (viz., the conic which exists in the general case will break up into a line-pair consisting of the line IJ and another line). The line in question may be called the focal axis; it will presently be shown that in the case of the circular cubic it passes through the centre, and that in the case of the bicircular quartic it not only passes through the centre, but coincides with one or other of the nodal axes, viz., with that passing through the real or the imaginary nodo-foci; that is, the curve may have on the focal axis two real or else two imaginary nodo-foci. The focal axis contains, as has been mentioned, four foci; the remaining twelve foci are situate symmetrically, six on each side of the focal axis, the arrangement of the sixteen foci being as mentioned ante, No. 81 *et seq.*; the focal axis is in fact an axis of symmetry of the curve, and if preferred it may be named the axis of symmetry, transverse axis, or simply the axis. And the curve (circular cubic, or bicircular quartic) is in this case a "symmetrical" or "axial" curve.

Article Nos. 131 to 140. Circular Cubic and Bicircular Quartic: Singular Forms.

131. The circular cubic may have a node or a cusp. If this were at one of the points I, J the curve would be imaginary, and I do not attend to the case; and for the same reason, for the bicircular quartic I do not attend to the case where one of

132. I consider first the case of the bicircular quartic where each of the points I, J is a cusp. The curve is in this case of necessity symmetrical[1]; it is in fact a Cartesian; viz., the Cartesian may be taken by definition to be a quartic curve having a cusp at each of the circular points at infinity. But in this case, as distinguished from the general case of the bicircular quartic, there is an essential degeneration of all the focal properties, and it is necessary to explain what these become. The centre is evidently the intersection of the cuspidal tangents; the nodo-foci (so far as they can be said to exist) coalesce with the centre, and they do not in so coalescing determine any definite directions for the nodal axes; that is, there are no nodal axes, and the only theorem in regard to the focal axis or axis of symmetry is, that it passes through the centre. Of the four tangents through the point I, one has come to coincide with the line IJ; and similarly, of the four tangents through the point J one has come to coincide with the line JI: there remain only three tangents through I and three tangents through J, and these by their intersections determine nine foci, viz., three foci A, B, C on the axis, and besides (B_1, C_1) the antipoints of (B, C); (C_1, A_1) the antipoints of (C, A); and (A_1, B_1) the antipoints of (A, B).

133. The remaining seven foci have disappeared, viz., we may consider that one of them has gone off to infinity on the focal axis, and that three pairs of foci have come to coincide with the points I, J respectively. The circle (as in the general case of a symmetrical quartic) has become a line, the focal axis; the circles R, S, T (contrary to what might at first sight appear) continue to be determinate circles, viz., these have their centres at A, B, C respectively, and pass through the points (B_1, C_1), (C_1, A_1), and (A_1, B_1) respectively, see ante, No. 83. But on each of these circles we have not more than two proper foci, and it is only on the axis as representing the circle that we have three proper foci, the axial foci A, B, C: in regard hereto it is to be remarked that the equation of the curve can be expressed not only by means of these three foci in the form $\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} = 0$; but by means of any two of them in the form $\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + K = 0$, where K is a constant, or, what is the same thing (z being introduced for homogeneity in the expressions of $\mathfrak{A}$ and $\mathfrak{B}$ respectively), in the form $\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + Kz^2 = 0$.

134. Using for the moment the expression "twisted" as opposed to symmetrical (viz., the curve is twisted when there is not any axis of symmetry, but the foci lie only on circles) then the classification is

Circular Cubics, twisted,
symmetrical,
Bicircular Quartics, twisted,

$$\text{symmetrical,} \left\{ \begin{array}{l} \text{Ordinary,} \\ \text{Bicuspidal} = \text{Cartesian,} \end{array} \right.$$

and each of these kinds may be general, nodal, or cuspidal; viz., for the two last mentioned kinds there may be a node or a cusp at a real point of the curve.

[1]It will appear, post Nos. 161—164, that if starting with three given points as the foci of a bicircular quartic, we impose the condition that the nodes at I, J shall be each of them a cusp, then either the quartic will be the circle through the three points taken twice, in which case the assumed focal property of the given three points disappears altogether, or else the three points must be in lined, and thus the curve be symmetrical, that is, a Cartesian.

135. In the case of a node,—say the point N; first if the curve (circular cubic or bicircular quartic) be twisted then of the four foci A, B, C, D we have two, suppose B and C, coinciding with N; and the sixteen foci are as follows, viz.

B, C, A, D are N, N, A, D;		
B_1, C_1, A_1, D_1	„	N, N, Antipoints of (A, D);
C_1, A_1, B_1, D_1	„	Antipoints of (N, A), Antipoints of (N, B);
A_3, B_3, C_3, D_3	do.	

viz., we have the points (A, D) each once, the node four times, the antipoints of (A, D) once, and the antipoints of (N, A) and of (N, D), each pair twice. But properly there are only four foci, viz., the points A, D and their antipoints. The circle subsists as in the general case, and so does the circle $R(BC, AD)$, viz., this has for centre the intersection of the line AB by the tangent at N to the circle C, and it passes through the point N, of course cutting the circle at right angles; the circles S and T each reduce themselves each to the point N considered as an evanescent circle, or what is the same thing to the line-pair NI, NJ.

136. The case is nearly the same if the curve be symmetrical, but in the case of the bicircular quartic excluding the Cartesian: viz., we have on the axis the foci B, C coinciding at N, and the other two foci A, D; the sixteen foci are as above and the circle R is determined by the proper construction as applied to the case in hand, viz., the centre is the intersection of the axis by the radical axis of the point N (considered as an evanescent circle) and the circle on AD as diameter; that is $RN^2 = RA \cdot RB$. And the circles S and T reduce themselves each to the point N considered as an evanescent circle.

137. Next if we have a cusp, say the point K: first if the curve (circular cubic or bicircular quartic) be twisted then of the four foci A, B, C, D, three, suppose A, B, C, coincide with K; and the sixteen foci are as follows, viz.,

B, C, A, D are K, K, K, D;		
B_1, C_1, A_1, D_1	„	K, K, Antipoints of (K, D),
C_1, A_1, B_1, D_1	do.	
A_3, A_2, C_1, D_3	do.	

viz., we have the point D once, the point K nine times, and the antipoints of K, D three times. But properly the point D is the only focus. The circle is, it would appear, any circle through K, D, but possibly the particular circle which touches the cuspidal tangent may be a better representative of the circle of the general case; the circles R, S, T reduce themselves each to the point K considered as an evanescent point.

138. The like is the case if the curve be symmetrical, but in the case of the bicircular quartic excluding the Cartesian; the circle is here the axis, which is in fact the cuspidal tangent.

139. For the Cartesian, if there is a node N; then of the three foci A, B, C, two, suppose B and C, coincide with N; the nine foci are A once, N four times, and the antipoints of N, A twice; but properly the point A is the only focus. And if there be a cusp K; then all the three foci A, B, C coincide with K; and the nine foci are K nine times; but properly the point K is the only focus.

140. The conclusion is, that in the singular forms the focal properties undergo an essential degradation, and that we have in the several cases only a limited or even no proper set of foci.

Article Nos. 141 to 149. Analytical Theory for the Circular Cubic.

141. The equation for the circular cubic may be taken to be

$$\xi(\eta^2 + \theta\eta z + ez^2) + z^2(a\xi + b\eta + cz) = 0,$$

viz. (ξ, η, z) being any coordinates whatever, this is the general equation of a cubic passing through the points $(\xi = 0, z = 0)$, $(\eta = 0, z = 0)$, and at these points touched by the lines $\xi = 0$, $\eta = 0$ respectively. And if $(\xi, \eta, z = 1)$ be circular coordinates, then we have the general equation of a circular cubic having the lines $\xi = 0$, $\eta = 0$ for its asymptotes, or say the point $\xi = 0$, $\eta = 0$ for its centre; the equation of the remaining asymptote is evidently $p\xi + q\eta + ez = 0$; to make the curve real we must have (p, q) and (a, b) conjugate imaginaries, e and c real.

142. Taking in any case the points I, J to be the points $\xi = 0, z = 0$ and $\eta = 0, z = 0$ respectively, for the equation of a tangent from I write $p\xi = \theta z$; then we have

$$\theta\eta(\theta z + q\eta + ez) + z(a\theta z + bp\eta + cpz) = 0,$$

that is

$$z^2(a\theta + cp) + \eta z(\theta^2 + e\theta + bp) + \eta^2 \cdot q\theta = 0,$$

and the line will be a tangent if only

$$(\theta^2 + e\theta + bp)^2 - 4q\theta(a\theta + cp) = 0,$$

that is

$$\theta^4 + 2e\theta^3 + \theta^2(e^2 + 2bp - 4aq) + 2\theta(bp - 2cq) + b^2p^2 = 0,$$

which is the equation for the determination of the tangents from I.

143. The first and second cases are those in which two roots are equal, giving a node at I or a cusp at I respectively; and similarly the third and fourth cases are those in which two roots are equal, giving a node at J or a cusp at J respectively. For a node at I we have $bp = 0$ and $e = 0$; but if $p = 0$, then the cubic is $(\xi\eta + z^2)(b\eta + cz) + e\eta^2 z = 0$, which has a cusp at J; hence for a node at I we must have $b = 0$, $e = 0$, and the equation is

$$\xi(\eta^2 + \theta\eta z) + z^2(a\xi + cz) = 0.$$

For a cusp at I we have the same conditions $b = 0$, $e = 0$, and the further condition $aq = 0$; but if $a = 0$, then the cubic is $(\xi\eta^2 + q\xi\eta z + cz^3) = 0$, which is a line and a conic; hence we must have $q = 0$, and the equation is

$$\xi\eta^2 + z^2(a\xi + cz) = 0.$$

Similarly for a node or cusp at J we have $a = 0$, $e = 0$, with $q = 0$ for the cusp, and the equations are

$$\eta(\xi^2 + p\xi z) + z^2(b\eta + cz) = 0 \quad \text{and} \quad \eta\xi^2 + z^2(b\eta + cz) = 0.$$

144. First if $p = q$, $a = b$: the equation becomes

$$\xi\eta(\eta + pz) + z^2(a\xi + a\eta + cz) = 0,$$

which has a node at the point $(\xi = 0, \eta = 0)$, that is, the centre.

145. The equation for the bicircular quartic may be taken to be

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2\{a\xi + b\eta + cz^2 + \xi\eta(a'\xi + b'\eta) + c'z^2\} = 0,$$

viz. (ξ, η, z) being any coordinates whatever, this is the equation of a quartic curve having the points $(\xi = 0, z = 0)$ and $(\eta = 0, z = 0)$ for nodes. And if $(\xi, \eta, z = 1)$ be circular coordinates, then we have the equation of a bicircular quartic, the points I, J being the nodes.
146. The equation may be written

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2 \cdot u = 0,$$

where $u = 0$ represents a conic. For the tangents from I we have $\xi^2 = \alpha z^2$; the equation gives $\xi^2 - \alpha z^2 = 0$, and thence $k\alpha(\eta^2 - \beta z^2) + u = 0$; that is,

$$\eta^2(k\alpha + b') + \eta(b\xi + ez) + k\alpha(a - \beta)z^2 + a\xi^2 + c\xi z + c'z^2 = 0.$$

But writing herein $\xi^2 = \alpha z^2$, $\xi = \sqrt{\alpha} \cdot z$, this becomes

$$\eta^2(k\alpha + b') + \eta(b\sqrt{\alpha} \cdot z + ez) + z^2(k\alpha^2 + a\alpha + c\sqrt{\alpha} + c') = 0,$$

which is a quadratic for the determination of the tangents from I; viz., these are $\xi = \sqrt{\alpha} \cdot z$, where $\sqrt{\alpha}$ is a root of

$$k\alpha^2 + (a + b'\beta)\alpha + c\sqrt{\alpha} + c' + e\beta = 0.$$

Similarly for the tangents from J.

147. The two equations may be written

$$24k\alpha^2\theta = 8\beta\theta^3 + 8k\alpha\beta\cdot\theta - Gka\alpha\theta - 8k^2\alpha\beta\cdot\theta - 4kc\theta + e^2,$$
$$-8k^2\alpha\beta\cdot\theta - 4kc\theta + e^2,$$
$$10\theta^2 - 6\alpha\alpha + 3e - 6kb'\beta + 3e'$$
$$24k\alpha^2\beta^2 + 24kc + 6$$

which equations have the same invariants; in fact for the first equation the invariants are found to be as follows, viz., if for shortness $\Theta = -8k^2\alpha\beta\cdot 8k^2 - 4kc\theta + e^2$, then

$$I = \Theta\{576k^2\alpha^2\beta^2 + 576k^2c\alpha\beta^2 + 144k^2(a^2\alpha + b'^2\beta) + 72k\cdot e\alpha\beta + 3c^2\}.$$

$$J = \Theta\{576k^2\alpha^2\beta^2 + 576k^2c\alpha\beta^2 + 144k^2(a^2\alpha + b'^2\beta) + 36k\cdot e\alpha\beta - c^3\}$$
$$-c^2\{576k^2\alpha\beta\cdot c\alpha\beta^2 + 144k^2c(a^2\alpha + b'^2\beta) + 36k\cdot e\alpha\beta^2\}$$

and then by symmetry the other equation has the same invariants. The absolute invariant $I^3 \div J^2$ has thus the same value in the two equations, that is, the equations are linearly transformable the one into the other.

148. In the first case, if θ be determined by the foregoing equation, we may take as corresponding tangents through the two nodes respectively $\xi = \sqrt{\alpha}\cdot z$, $\eta = \sqrt{\beta}\cdot z$; the foci (A, B, C, D), which are the intersections of the pairs of lines $(\xi = \sqrt{\alpha}\cdot z, \eta = \sqrt{\beta}\cdot z)$, &c., lie, it is clear, in the line $\beta\xi - \alpha\eta = 0$, which is one of the nodal axes of the curve. Similarly, in the second case, if θ be determined by the foregoing equation, we may take as corresponding tangents through the two nodes respectively $\xi = \sqrt{\alpha}\cdot z$, $\eta = -\sqrt{\beta}\cdot z$; the foci (A, B, C, D), which are the intersections of the pairs of lines $(\xi = \sqrt{\alpha}\cdot z, \eta = -\sqrt{\beta}\cdot z)$, &c., lie in the line $\beta\xi + \alpha\eta = 0$, which is the other of the nodal axes of the curve. In either case the foci A, B, C, D lie in a line, that is, we have the curve symmetrical; and, as we have just seen, the focal axis, or axis of symmetry, is one or other of the nodal axes.

149. In the case of the Cartesian, or when $\alpha = 0$, $\beta = 0$, viz., the equation $a\alpha = b\beta$ is satisfied identically, and this seems to show that the Cartesian is symmetrical; it is to be observed, however, that for $\alpha = 0$, $\beta = 0$ the foregoing formulæ fail, and it is proper to repeat the investigation for the special case.

Article Nos. 150 to 153. Casey's Theorem for the Bicircular Quartic.

150. The theorem that, for a bicircular quartic, if $(\xi - \alpha z = 0, \eta - \alpha' z = 0)$, $(\xi - \beta z = 0, \eta - \beta' z = 0)$, $(\xi - \gamma z = 0, \eta - \gamma' z = 0)$, $(\xi - \delta z = 0, \eta - \delta' z = 0)$ are the equations of the pairs of tangents from the four foci A, B, C, D respectively, then the equation of the curve is

$$\sqrt{(\xi - \alpha z)(\eta - \alpha' z)} + \sqrt{(\xi - \beta z)(\eta - \beta' z)} + \sqrt{(\xi - \gamma z)(\eta - \gamma' z)} + \sqrt{(\xi - \delta z)(\eta - \delta' z)} = 0,$$

where the radicals are connected by a linear equation, may be established as follows.

151. Consider the case of the bicircular quartic, and take as before $(\xi = 0, z = 0)$ and $(\eta = 0, z = 0)$ for the coordinates of the points I, J respectively. Let the two tangents from the focus A be $\xi - \alpha z = 0$, $\eta - \alpha' z = 0$, say for shortness $p = 0$, $p' = 0$, then the equation of the curve is expressible in the form $pp'\Omega = \mho^2$, where $\Omega = 0$, $\mho = 0$ are each of them a circle, viz., Ω and $\mho$ are each of them a quadric function containing the terms z^2, $z\eta$, $z\xi$, and $\xi\eta$. Taking an indeterminate coefficient λ, the equation may be written

$$pp'(\Omega + 2\lambda\mho + \lambda^2 pp') = (\mho + \lambda pp')^2,$$

and then λ may be so determined that $\Omega + 2\lambda\mho + \lambda^2 pp' = 0$, shall be a 0-circle, that is, shall have its centre at I; viz. this will be the case if the coefficients of η^2, ηz, z^2 are each $= 0$; or, what is the same thing, if the circle reduces itself to the line-pair $\xi = 0$, $z = 0$; that is, if the coefficients of ξ^2, $\xi\eta$, ξz, η^2, ηz, z^2 are proportional to $0, 0, 0, 0, 0, 1$ respectively.

152. We have to show that the four foci $(p = 0, p' = 0)$, $(q = 0, q' = 0)$, $(r = 0, r' = 0)$, $(s = 0, s' = 0)$, lie in a line; that is, that the line in question is an axis of the curve. But this is a particular case of the theorem that, if $(p = 0, p' = 0)$, $(q = 0, q' = 0)$, $(r = 0, r' = 0)$ are any three foci, then the circle through the three foci (which, it has been shown, exists for any three foci) is the circle R. In fact, if the three foci lie in a line, then the circle R is the line, or say the circle on the line at infinity between the two circular points at infinity; that is, the circle is the line at infinity, and the axis is thus shown to be a line containing the four foci.

153. The analytical investigation of the focal properties of the bicircular quartic may be presented under a somewhat different form. Writing the equation as

$$(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + (a\xi + b\eta + cz^2 + a'\xi\eta + b'\xi^2 + c'\eta^2)z^2 = 0,$$

the condition that this shall represent a bicircular quartic is that α and β shall each be determinate, so that the nodes at I and J shall be proper nodes; and this requires that k shall not $= 0$, and that a', b' shall not be both of them $= 0$. If $k = 0$, the curve degenerates into a conic twice; if $a' = 0$, $b' = 0$, the curve degenerates into a pair of circles.

154. The foregoing conclusions may be verified by means of the expressions for the foci in terms of the coefficients of the equation. For if we write

$$\Omega = a\xi + b\eta + cz^2 + a'\xi\eta + b'\xi^2 + c'\eta^2,$$

then the equation is

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2\Omega = 0,$$

and for the tangents at $(\xi = 0, z = 0)$ we have $\xi^2 = \alpha z^2$, and thence

$$k\alpha(\eta^2 - \beta z^2) + \Omega = 0;$$

similarly for finding the tangents at $(\eta = 0, z = 0)$ we have only to to interchange ξ and η, α and β, and write Ω' in place of Ω.

155. The condition for the determination of the foci is that the equation

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} + \sqrt{\mathfrak{D}} = 0$$

shall be expressible as a determinant; viz. this is

$$\begin{vmatrix} \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \sqrt{\mathfrak{D}} \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0,$$

where the equation is satisfied by writing

$$\sqrt{\mathfrak{A}} : \sqrt{\mathfrak{B}} : \sqrt{\mathfrak{C}} : \sqrt{\mathfrak{D}} = \lambda : \mu : \nu : \rho,$$

where $\lambda + \mu + \nu + \rho = 0$.

156. Calculating $AB' - A'B$, $CA' - C'A$, $CD' - CD$, $BD' - B'D$, these are at once found to be proportional to a, b, c, d respectively; and hence we have the theorem that for a bicircular quartic if $(\xi - \alpha z = 0, \eta - \alpha' z = 0)$ are the equations of the tangents from any focus, then the equation of the curve may be written in the form

$$a\sqrt{(\xi-\alpha z)(\eta-\alpha' z)} + b\sqrt{(\xi-\beta z)(\eta-\beta' z)} + c\sqrt{(\xi-\gamma z)(\eta-\gamma' z)} + d\sqrt{(\xi-\delta z)(\eta-\delta' z)} = 0,$$

where a, b, c, d are proportional to the distances of the foci from any line whatever.

Article Nos. 157 to 160. The Focal Conic.

157. I attend in the first place to the focal conic of the bicircular quartic; the theorem is that given any three circles A, B, C, there exists a conic passing through the centres of these three circles, and having the property that, taking on the conic any point D as the centre of a fourth circle D, the four circles have a common orthogonal circle; or, what is the same thing, that the four circles are tangential to a fifth circle. The proof may be presented as follows.

158. Let the equations of the three circles be

$$x^2+y^2+2gx+2fy+c=0, \quad (A), x^2+y^2+2g'x+2f'y+c'=0, \quad (B), x^2+y^2+2g''x+2f''y+c''=0, \quad (C),$$

then the equation of any circle orthogonal to these three is

$$\begin{vmatrix} x^2+y^2 & x & y & 1 \\ -c & g & f & -1 \\ -c' & g' & f' & -1 \\ -c'' & g'' & f'' & -1 \end{vmatrix} = 0;$$

and the condition that a fourth circle

$$x^2+y^2+2G\xi+2F\eta+C=0,$$

shall be orthogonal to this, is found to be

$$\lambda(aG+bF+C+d) + \mu(a'G+b'F+C+d') + \nu(a''G+b''F+C+d'') = 0,$$

which is the equation of a conic passing through the centres of the three circles A, B, C.

159. To find the envelope of a circle whose centre lies on a given conic, and which cuts at right angles a given circle. Let the conic be

$$(a, b, c, f, g, h|x, y, z)^2 = 0,$$

and the circle be $(x-\alpha)^2+(y-\beta)^2 = r^2$; then for the circle through the centre (x, y, z) we have

$$(X-x)^2 + (Y-y)^2 = \rho^2,$$

where $\rho^2 = (x-\alpha)^2 + (y-\beta)^2 - r^2$. The condition of orthogonality gives

$$x^2 + y^2 - \rho^2 = 0, \quad \text{or} \quad x^2 + y^2 - (x-\alpha)^2 - (y-\beta)^2 + r^2 = 0,$$

that is

$$2\alpha x + 2\beta y - (\alpha^2 + \beta^2 - r^2) = 0,$$

which is a line; and the envelope is found by combining this with the conic.

160. In particular, if the conic is a conic passing through the points I, J, then the envelope is a bicircular quartic. For if the conic passes through I and J, then the condition that the centre of the variable circle shall lie on the conic is

$$l\xi + m\eta + n + \lambda\sqrt{(\xi\eta)} = 0,$$

and the envelope of the circle is found to be

$$(l^2, m^2, n^2, -mn, -nl, -lm|\xi^2, \eta^2, z^2, \eta z, z\xi, \xi\eta) = 0,$$

which is the equation of a bicircular quartic.

Article Nos. 161 to 164. Condition that a Bicircular Quartic shall be a Cartesian.

161. The condition that a bicircular quartic shall be a Cartesian, or shall have each of its nodes at I and J a cusp, may be investigated as follows. The equation is

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2\Omega = 0,$$

and for a cusp at I we must have $\alpha = 0$ and the tangent $\xi = 0$ a cuspidal tangent; that is, writing $\xi = 0$ in the equation, the resulting equation in (η, z) must have a cusp at $(\eta = 0, z = 0)$, which requires that the terms in η^4, $\eta^3 z$ shall vanish, and that the term $\eta^2 z^2$ shall not vanish. The conditions are thus found to be $k\beta + c' = 0$ and $e = 0$; and similarly for a cusp at J, $k\alpha + b' = 0$ and $e' = 0$.

162. We have thus the conditions for the Cartesian

$$b' = -k\alpha, \quad c' = -k\beta, \quad e = 0, \quad e' = 0,$$

and the equation becomes

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2(a\xi + b\eta + cz^2 - k\alpha\eta^2 - k\beta\xi^2) = 0,$$

which may be written

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) - k\alpha z^2\eta^2 - k\beta z^2\xi^2 + z^2(a\xi + b\eta + cz^2) = 0,$$

or, what is the same thing,

$$k(\xi^2\eta^2 - \alpha\eta^2 z^2 - \beta\xi^2 z^2 + \alpha\beta z^4) - k\alpha z^2\eta^2 - k\beta z^2\xi^2 + z^2(a\xi + b\eta + cz^2) = 0,$$

that is

$$k\xi^2\eta^2 - 2k\alpha\eta^2 z^2 - 2k\beta\xi^2 z^2 + k\alpha\beta z^4 + z^2(a\xi + b\eta + cz^2) = 0.$$

But this may be reduced to the standard form of the Cartesian.

163. The conditions may also be expressed in terms of the foci. If the foci are given as the intersections of the pairs of lines $(\xi = \alpha_r z, \eta = \beta_r z)$, then the condition is that the four pairs of lines shall be such that the product equation

$$\prod_{r=1}^{4}(\eta - \beta_r z) = 0$$

shall have the coefficients of η^4, $\eta^3 z$ each $= 0$; and similarly for the other pair of tangents.

164. In particular, if three foci A, B, C are given, then for the Cartesian the fourth focus D is determined as the intersection of two known lines; and it appears that the three foci must lie on a line, that is, the Cartesian is necessarily symmetrical.

165. There is not, in general, any identical equation

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

but if the four foci lie on a circle, then such an equation exists. In fact, if the foci lie on a circle, then we may write

$$\sqrt{\mathfrak{A}} : \sqrt{\mathfrak{B}} : \sqrt{\mathfrak{C}} : \sqrt{\mathfrak{D}} = \lambda : \mu : \nu : \rho,$$

where $\lambda + \mu + \nu + \rho = 0$, and the condition is satisfied.

Article Nos. 166 to 170. Applications of the Focal Theory.

166. The focal theory of the bicircular quartic leads to many interesting applications. In particular, it gives a construction for the quartic when the four foci and the constant k are given; and it leads to the investigation of the bitangents and of the nodes and cusps of the curve.

167. If the four foci are given, then the equation of the curve is at once written down in the form

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} + \sqrt{\mathfrak{D}} = 0,$$

where $\mathfrak{A} = 0$, $\mathfrak{B} = 0$, $\mathfrak{C} = 0$, $\mathfrak{D} = 0$ are the equations of the pairs of tangents from the four foci respectively.

168. The equation may be rationalized by writing it in the form

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = -\sqrt{\mathfrak{C}} - \sqrt{\mathfrak{D}},$$

and squaring; we thus obtain

$$\mathfrak{A} + \mathfrak{B} + 2\sqrt{\mathfrak{A}\mathfrak{B}} = \mathfrak{C} + \mathfrak{D} + 2\sqrt{\mathfrak{C}\mathfrak{D}},$$

that is

$$\mathfrak{A} + \mathfrak{B} - \mathfrak{C} - \mathfrak{D} = 2(\sqrt{\mathfrak{C}\mathfrak{D}} - \sqrt{\mathfrak{A}\mathfrak{B}}),$$

and squaring again,

$$(\mathfrak{A} + \mathfrak{B} - \mathfrak{C} - \mathfrak{D})^2 = 4(\mathfrak{C}\mathfrak{D} + \mathfrak{A}\mathfrak{B} - 2\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D}}).$$

The rationalized equation is thus of the fourth order in (ξ, η, z), and represents a bicircular quartic.

169. The bitangents of the bicircular quartic may be investigated by means of the focal equation. A bitangent is a line which touches the curve in two points; and if the line be taken as $lx + my + nz = 0$, then the condition of tangency gives a relation between the coefficients (l, m, n). It appears that there are, in general, four bitangents, and that these correspond to the four foci in such wise that each bitangent is the polar of a focus with respect to a certain conic.

170. The investigation leads to the consideration of the Jacobian of four circles, and to the theorem that the four foci of a bicircular quartic are the centres of the four circles which are orthogonal to three given circles. The theory is intimately connected with that of the "focal conic" already referred to.

Article Nos. 171 and 172. Locus of Centre of Variation.

171. The equation of the bicircular quartic may be written in the form

$$\frac{\xi^2}{a} + \frac{\eta^2}{b} + \frac{z^2}{c} + \frac{2\eta z}{a'} + \frac{2z\xi}{b'} + \frac{2\xi\eta}{c'} = 0,$$

and the centre of the curve is then determined as the intersection of the lines

$$\frac{\xi}{a} + \frac{\eta}{c'} + \frac{z}{b'} = 0, \quad \frac{\xi}{c'} + \frac{\eta}{b} + \frac{z}{a'} = 0.$$

172. If the curve varies, the locus of the centre is found by eliminating the variable parameters between the equations; and it appears that the locus is, in general, a conic. This conic is the "focal conic" of the system of bicircular quartics.

173. The line of centres is the axis of the curve, but the centres A, B, C, D are not in general on a line; they are the centres of the circles A, B, C, D respectively, and the condition that the four circles shall have a common orthogonal circle is the condition already referred to in connexion with the focal conic.

Article Nos. 174 to 177. Casey's Equation.

174. Casey's well-known equation for four circles touching a fifth circle may be established by means of the focal properties of the bicircular quartic. If four circles touch a fifth circle, then the distances of their centres from any line satisfy a certain linear relation; and conversely, if such a relation exists, the four circles touch a fifth circle.

175. Let the four circles be A, B, C, D, and let their equations be

$$x^2 + y^2 + 2g_r x + 2f_r y + c_r = 0, \quad (r = 1, 2, 3, 4).$$

If these touch a fifth circle, then writing the equation of the fifth circle as

$$x^2 + y^2 + 2Gx + 2Fy + C = 0,$$

the condition of tangency gives

$$(G - g_r)^2 + (F - f_r)^2 = (C - c_r)^2,$$

for $r = 1, 2, 3, 4$. These are four equations which must be consistent, and the condition of consistency is Casey's equation.

176. Casey's equation may be written in the form

$$\begin{vmatrix} d_{12} & d_{13} & d_{14} \\ d_{21} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{34} \end{vmatrix} = 0,$$

where d_{rs} is the distance between the centres of the circles r and s, measured with the proper sign. The equation is symmetrical in regard to the four circles, and it is the condition that the four circles shall be tangential to a fifth circle.

177. The equation may also be presented in a form which brings into view the connexion with the bicircular quartic. If the four circles have a common orthogonal circle, then the four foci lie on a conic; and conversely, if the four foci lie on a conic, then the four circles have a common orthogonal circle. This is the theorem of the focal conic already referred to.

Article Nos. 178 to 181. Case of Double Contact; Casey's Equation in the Problem of the Tangent Circles.

178. In the problem of the tangent circles,—given three circles, to draw the circles touching all three,—there are, in general, eight solutions. The investigation leads to Casey's equation, which gives the condition that four circles shall touch a fifth. The eight circles may be divided into four sets of two, each set touching a different one of the four pairs of circles which are the inverses of the given three circles.

179. The equation for the tangent circles may be obtained as follows. Let the three given circles be A, B, C, and let the required circle be S. Let d, d', d'' be the distances of the centre of S from the centres of A, B, C respectively, and let r, r', r'', ρ be the radii of A, B, C, S respectively. Then for external contact we have

$$d = r + \rho, \quad d' = r' + \rho, \quad d'' = r'' + \rho,$$

and for internal contact the corresponding equations with the signs of r, r', r'' changed. The elimination of ρ between these equations leads to a relation between d, d', d'' which is the condition that the circle S shall touch the three circles A, B, C.

180. The condition may be expressed as the vanishing of a determinant. Writing the equations

$$d - r - \rho = 0, \quad d' - r' - \rho = 0, \quad d'' - r'' - \rho = 0,$$

and eliminating ρ, we obtain

$$\begin{vmatrix} 1 & 1 & 0 & 0 \\ r & 0 & d & -1 \\ r' & d' & 0 & -1 \\ r'' & 0 & d'' & -1 \end{vmatrix} = 0,$$

which is the condition for the tangent circle. This condition is equivalent to Casey's equation for the four circles A, B, C, S.

181. The investigation shows that the problem of the tangent circles is intimately connected with the theory of the bicircular quartic. The eight tangent circles correspond to the eight foci of a certain bicircular quartic, and the conditions of tangency are given by the focal equation of this quartic.

Article Nos. 182 and 183. Focal Formulæ for the General Curve.

182. Comparing the formula of No. 120 with that of No. 50, we see that the focal formulae for the general curve are as follows:—If the coordinates of a point on the curve are proportional to (ξ, η, ζ), and if (α, β, γ), $(\alpha', \beta', \gamma')$, &c. are the coordinates of the foci, then the distances of the point from the foci are proportional to

$$\sqrt{(a\xi + b\eta + c\zeta)(a'\xi + b'\eta + c'\zeta)},$$

and the equation of the curve is obtained by equating to zero a linear function of the square roots of these expressions.

183. In the case of a curve of the nth order, there are n^2 foci, and the equation of the curve may be written in the form

$$\sum_{r=1}^{n^2} \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r = 0$ is the equation of the pair of tangents from the rth focus. This is the generalization of the focal property of the conic, and it applies to all algebraic curves.

Article Nos. 184 and 185. Case of the Circular Cubic.

184. In the case of the circular cubic, the equation may be written in the form

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} = 0,$$

where $\mathfrak{A} = 0$, $\mathfrak{B} = 0$, $\mathfrak{C} = 0$ are the equations of the pairs of tangents from the three foci respectively. The curve has three foci, and the equation is the condition that the sum of the distances from the three foci, measured in a certain way, shall be zero.

185. The rationalized equation of the circular cubic is obtained by squaring twice, and it is found to be of the form

$$(\mathfrak{A} + \mathfrak{B} + \mathfrak{C})^2 - 4(\mathfrak{A}\mathfrak{B} + \mathfrak{B}\mathfrak{C} + \mathfrak{C}\mathfrak{A}) = 0,$$

which is a particular case of the general equation of the third order. The curve has a node at each of the circular points at infinity, and the three foci lie on a line, which is the axis of the curve.

186. For the circular cubic, the three foci A, B, C lie on a line, and the equation of the curve may be written in the form

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} = 0,$$

where a, b, c are constants. The rationalized equation is

$$a^2\mathfrak{A} + b^2\mathfrak{B} + c^2\mathfrak{C} + 2bc\sqrt{\mathfrak{B}\mathfrak{C}} + 2ca\sqrt{\mathfrak{C}\mathfrak{A}} + 2ab\sqrt{\mathfrak{A}\mathfrak{B}} = 0,$$

and squaring again we obtain the equation of the cubic.
187. The focal circles of the circular cubic are three circles, each passing through two of the foci and through the node at infinity. The centres of these circles lie on the axis of the curve, and they are the centres of the three pairs of antipoints of the foci.
188. The classification of circular cubics is as follows:—

(1) General circular cubic: three distinct foci, no singularities.

(2) Nodal circular cubic: a node at a finite point, two of the foci coinciding at the node.

(3) Cuspidal circular cubic: a cusp at a finite point, all three foci coinciding at the cusp.

(4) Symmetrical circular cubic: the three foci on a line, the curve having an axis of symmetry.

Each of these may be real or imaginary, and the curve may have one or two branches.
189. These values of $\sqrt{l}$, $\sqrt{m}$, $\sqrt{n}$ give the equations of the two circles through the three foci; and the envelope of the line joining the points of contact of tangent circles is the focal axis of the curve.

190. The theorem that the four foci of a bicircular quartic lie on a circle may be proved directly from the focal equation. If the equation is

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} + \sqrt{\mathfrak{D}} = 0,$$

then the condition that the four foci shall lie on a circle is that there shall exist a linear relation between the square roots; and this condition is satisfied identically when the four foci lie on a circle.

191. The converse theorem is also true: if the four foci lie on a circle, then the curve is a bicircular quartic. This follows from the fact that the focal equation determines the curve uniquely when the foci and the constant k are given.

192. The focal circles of the bicircular quartic are four circles, each passing through three of the foci. The centres of these circles lie on the focal conic, and the circles are orthogonal to the common orthogonal circle of the four foci.

193. The eight foci of a bicircular quartic consist of four real and four imaginary points, or of two real and six imaginary points, or of eight imaginary points, according to the nature of the curve. The real foci lie on the real focal circles, and the imaginary foci on the imaginary focal circles.

194. The sixteen foci of the general bicircular quartic are arranged in four tetrads of concyclic foci. Each tetrad gives rise to a focal circle, and the four focal circles have a common orthogonal circle. The centres of the four focal circles lie on the focal conic.

195. The focal properties of the bicircular quartic are closely connected with the theory of the bitangents. Each bitangent of the quartic corresponds to a pair of foci, and the four bitangents correspond to the four pairs of foci which can be formed from the four foci.
196. The nodes and cusps of the bicircular quartic may be investigated by means of the focal equation. A node corresponds to the coincidence of two foci, and a cusp to the coincidence of three foci. The conditions for a node or cusp may thus be expressed in terms of the focal coordinates.
197. The case of the Cartesian, where each of the circular points at infinity is a cusp, is the limiting case of the bicircular quartic when three of the foci coincide at each cusp. The focal equation then degenerates, and the nine foci reduce to three axial foci and their antipoints.
198. The focal theory may be extended to curves of higher order. A curve of order n has n^2 foci, and its equation may be written in the form

$$\sum_{r=1}^{n^2} \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r = 0$ is the equation of the pair of tangents from the rth focus. The theory of the focal circles and of the focal conic may be generalized accordingly.

Article Nos. 199 to 201. Tetrazomal Curves.

199. A tetrazomal curve is defined as the locus of a point such that the sum of the squares of the distances from four fixed points, each multiplied by a constant coefficient, is zero. If the four fixed points are A, B, C, D, and the coefficients are a, b, c, d, then the equation of the curve is

$$a \cdot PA^2 + b \cdot PB^2 + c \cdot PC^2 + d \cdot PD^2 = 0,$$

where P is any point on the curve.
200. The general equation of a tetrazomal curve may be written in the form

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are quadric functions of the coordinates. The curve is of the fourth order, and it has the four fixed points as foci.
201. The tetrazomal curve includes as particular cases the bicircular quartic (when the four fixed points are concyclic) and the circular cubic (when one of the coefficients is zero). The general theory of tetrazomal curves is thus a generalization of the focal theory of the bicircular quartic.

202. The classification of tetrazomal curves is as follows:—

I. General tetrazomal: four distinct foci, no singularities; order $= 4$.

II. Nodal tetrazomal: a node at one of the foci, or at another point; order $= 4$.

III. Cuspidal tetrazomal: a cusp at one of the foci, or at another point; order $= 4$.

IV. Degenerate tetrazomal: the curve breaks up into two conics, or into other simpler forms.

Each of these kinds may be real or imaginary, and the curve may have one or more branches.
203. The focal circles of a tetrazomal curve are circles passing through three of the four foci. There are four such circles, and they have a common orthogonal circle. The centres of the focal circles lie on a conic, which is the focal conic of the curve.
204. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; order of tetrazomal $= 4$; orders of trizomals $= 3$ each; corresponding to IV. supra.

Article Nos. 205 to 207. Pentazomal and Higher Curves.

205. V. $\sqrt{l}=\sqrt{m}$; order of tetrazomal $= 5$; orders of trizomals $= 3$ each.
206. The theory may be extended to pentazomal curves, defined by the equation

$$a\sqrt{\mathfrak{A}}+b\sqrt{\mathfrak{B}}+c\sqrt{\mathfrak{C}}+d\sqrt{\mathfrak{D}}+e\sqrt{\mathfrak{E}}=0,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$, $\mathfrak{E}$ are quadric functions of the coordinates. The curve is of the fifth order, and it has five foci.
207. VII. If we have $1,1,1,1=0$, and $(1,1,1,1)(\sqrt{l},\sqrt{m},\sqrt{n},\sqrt{p})=0$, the condition is satisfied identically; and conversely, if the condition is satisfied, the curve is a tetrazomal. This is the generalization of the theorem that four points lie on a circle if and only if the sum of the opposite angles of the quadrilateral is equal to two right angles.

208. The general theory of polyazomal curves includes all curves defined by an equation of the form

$$\sum_{r=1}^{n} a_r \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r$ are quadric functions of the coordinates. The curve is of order $2n$, and it has n foci. The theory of the focal circles, the focal conic, and the bitangents may be generalized to these curves.

209. The polyazomal curve of order $2n$ has n focal circles, each passing through $n-1$ of the n foci. The n focal circles have a common orthogonal circle, and their centres lie on a curve of order $n-1$, which is the focal curve of the polyazomal.

210. The bitangents of a polyazomal curve correspond to the pairs of foci, and their number is $\frac{1}{2}n(n-1)$. Each bitangent is the polar of a pair of foci with respect to a certain conic, and the envelope of the bitangents is a curve of order $2n-2$.

211. The conditions for a node or cusp of a polyazomal curve may be expressed in terms of the focal coordinates. A node corresponds to the coincidence of two foci, and a cusp to the coincidence of three foci. The conditions are analogous to those for the bicircular quartic.

212. The degenerate forms of polyazomal curves correspond to special relations between the foci and the coefficients. If two coefficients are equal, the curve has a node; if three are equal, it has a cusp; and if all are equal, it breaks up into simpler components.

213. The analytical theory of polyazomal curves may be developed by means of the method of symmetric functions. If the foci are given, and the coefficients are regarded as variables, then the equation of the curve may be written in the form

$$\sum_{r=1}^{n} a_r \sqrt{(\xi - \alpha_r z)(\eta - \beta_r z)} = 0,$$

and the rationalized equation is obtained by eliminating the radicals. This leads to an equation of order $2n$ in the coordinates.

214. The condition that a polyazomal curve shall have a node is that the discriminant of the rationalized equation shall vanish. This condition may be expressed as a determinant of order $2n$, whose elements are symmetric functions of the focal coordinates and the coefficients.

215. The condition that a polyazomal curve shall have a cusp is more complicated. It requires that the discriminant shall vanish, and also that certain subdeterminants shall vanish. The conditions may be expressed as the vanishing of the invariants of the rationalized equation.

216. The focal properties of polyazomal curves are closely connected with the theory of transformations. If a polyazomal curve is transformed by inversion, the foci are transformed into the foci of the transformed curve, and the coefficients are multiplied by certain factors. The focal circles and the focal curve are also transformed into the corresponding loci of the transformed curve.

217. The theory of polyazomal curves has many applications in geometry and physics. In geometry, it furnishes a general method for the investigation of curves with given focal properties. In physics, it is connected with the theory of potential, and with the problem of determining the equipotential surfaces of a system of charges.
218. The equation of a polyazomal curve may be interpreted as the condition that the sum of the distances from a variable point to a system of fixed points, each multiplied by a constant coefficient, shall be zero. This is a generalization of the definition of the ellipse and hyperbola as the locus of a point the sum or difference of whose distances from two fixed points is constant.
219. The investigation of the singularities of polyazomal curves is of great importance for the theory of algebraic curves. The results obtained for these curves may be extended, by means of birational transformation, to curves of the most general kind.
220. The memoir concludes with the indication of certain directions in which the theory may be extended. The theory of polyazomal curves is capable of generalization in many ways; and it is hoped that the present investigation may serve as a foundation for future researches on this interesting subject.

184. In the case of a circular cubic, we must have

$$\rho(\beta-\gamma)(\alpha-\delta)+\sigma(\gamma-\alpha)(\beta-\delta)+\tau(\alpha-\beta)(\gamma-\delta)=0,$$

$$\sqrt{\alpha\rho}(\beta-\gamma)+\sqrt{\beta\sigma}(\gamma-\alpha)+\sqrt{\gamma\tau}(\alpha-\beta)=0,$$

which, when the foci A, B, C, D are given, determine the values of $\rho:\sigma:\tau$ in order that the curve may be a circular cubic. We see at once that there are two sets of values, and consequently two circular cubics having each of them the given points A, B, C, D for a set of concyclic foci. The two systems may be written

$$\sqrt{\rho}:\sqrt{\sigma}:\sqrt{\tau}=\sqrt{\alpha\delta}-\sqrt{\beta\gamma}:\sqrt{\beta\delta}-\sqrt{\gamma\alpha}:\sqrt{\gamma\delta}-\sqrt{\alpha\beta},$$

viz., it being understood that $\sqrt{\alpha\delta}$ means $\sqrt{\alpha}\sqrt{\delta}$, &c., then, according as $\sqrt{\delta}$ has one or other of its two opposite values, we have one or other of the two systems of values of $\rho:\sigma:\tau$. To verify this, observe that the product

$$(\sqrt{\alpha\delta}-\sqrt{\beta\gamma})(\sqrt{\beta\delta}-\sqrt{\gamma\alpha})(\sqrt{\gamma\delta}-\sqrt{\alpha\beta})$$

is symmetrical in regard to α, β, γ, δ.

and for either set of values the verification of the relation

$$\sqrt{\alpha\rho}(\beta-\gamma)+\sqrt{\beta\sigma}(\gamma-\alpha)+\sqrt{\gamma\tau}(\alpha-\beta)=0,$$

will depend on the two identical equations

$$\sin A\sin(B-C)+\sin B\sin(C-A)+\sin C\sin(A-B)=0,$$

$$\cos A\sin(B-C)+\cos B\sin(C-A)+\cos C\sin(A-B)=0;$$

although the foregoing solution for the case of a circular cubic is the most elegant one, I will presently return to the question and give the solution in a different form.

Article No. 186. Focal Formulæ for the Symmetrical Curve.

186. In the symmetrical case, where the foci A, B, C, are on a line, then if, as usual, a, b, c, d denote the distances from a fixed point, we have the expressions of (a,b,c,d) in a form adapted to the formulae of No. 49, viz.,

$$a:b:c:d=(b-c)(c-d)(d-b):(c-d)(d-a)(a-c):(d-a)(a-b)(b-d):(a-b)(b-c)(c-a).$$

187. In fact, these values obviously satisfy the second equation; and to see that they satisfy the first equation, we have only to write them under the form

$$\rho : \sigma : \tau = M - \tfrac{1}{4}(b+c)(a+d) : M - \tfrac{1}{4}(c+a)(b+d) : M - \tfrac{1}{4}(a+b)(c+d),$$

where $M = (a+b+c+d)^2$. The first set gives for the curve

$$(b-c)\sqrt{\mathfrak{A}} + (c-a)\sqrt{\mathfrak{B}} + (a-b)\sqrt{\mathfrak{C}} = 0,$$

but this contains the line $z = 0$ not once only, but twice; it in fact is $(y^2 = 0)$, the axis taken twice; the only proper cubic with the foci A, B, C, in lined is therefore

$$(b-c)(a+d-b-c)\sqrt{\mathfrak{A}} + (c-a)(b+d-c-a)\sqrt{\mathfrak{B}} + (a-b)(c+d-a-b)\sqrt{\mathfrak{C}} = 0,$$

the equation of which is, of course, expressible in each of the other three forms.

Article Nos. 188 to 192. Case of the General Circular Cubic.

188. Returning to the general case of the circular cubic, the lines AD, BC meet in R, and if we denote by a_1, b_1, c_1, d_1 the distances from R of the four points respectively, so that $b_1c_1 = a_1d_1 = RA \cdot RD$ (observe that if as usual A, B, C, D are taken in order on the circle O, then A, C are on opposite sides of R, and B, D on opposite sides of R), then the two sets of values are

$$\sqrt{\rho} : \sqrt{\sigma} : \sqrt{\tau} = \sqrt{a_1d_1} - \sqrt{b_1c_1} : \sqrt{b_1d_1} - \sqrt{c_1a_1} : \sqrt{c_1d_1} - \sqrt{a_1b_1},$$

and

$$\sqrt{\rho} : \sqrt{\sigma} : \sqrt{\tau} = \sqrt{a_1d_1} + \sqrt{b_1c_1} : \sqrt{b_1d_1} + \sqrt{c_1a_1} : \sqrt{c_1d_1} + \sqrt{a_1b_1}.$$

189. These values of $\sqrt{l}$, $\sqrt{m}$, $\sqrt{n}$ give the equations of the two circular cubics with the foci (A, B, C, D), the equation of each of them under a fourfold form, viz., we have

$$(\cdot, d_1 - c_1, b_1 - d_1, c_1 - b_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$

$$(-d_1 + c_1, \cdot, a_1 - d_1, b_1 - a_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$
$$(-b_1 + d_1, -a_1 + d_1, \cdot, a_1 - b_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$
$$(-c_1 + b_1, -b_1 + a_1, -a_1 + c_1, \cdot)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$

(first curve), and

$$(\cdot, -c_1 - d_1, d_1 + b_1, -b_1 + c_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$

etc. (second curve).

190. Similarly AC and BD meet in S, and if we denote by a_2, b_2, c_2, d_2 the distances from S of the four points respectively, so that $c_2a_2 = b_2d_2 = SA \cdot SC$ (observe that if as usual A, B, C, D are taken in order on the circle O, then A, C are on opposite sides of S, and B, D on opposite sides of S), then we obtain the same two cubics, each equation under a fourfold form, with the values (a_2, b_2, c_2, d_2).

191. And similarly AB and CD meet in T, and if we denote by a_3, b_3, c_3, d_3 the distances from T of the four points respectively, so that $a_3b_3 = c_3d_3 = TA \cdot TB$, then we again obtain the same two cubics, each equation under a fourfold form, with the values (a_3, b_3, c_3, d_3).

192. The three systems have been obtained independently, but they may of course be derived each from any other of them: to show how this is, recollecting that we have

$$RA \cdot RD = RB \cdot RC, \quad SA \cdot SC = SB \cdot SD, \quad TA \cdot TB = TC \cdot TD,$$

these are the equations which connect the distances (a_1, b_1, c_1, d_1), (a_2, b_2, c_2, d_2), (a_3, b_3, c_3, d_3) respectively.

193. And hence

$$b_1(-b_1c_1 + c_1^2 + a_1d_1 - d_1^2) + c_1(-b_1d_1 + c_1d_1 + a_1d_1 - c_1d_1) = 0,$$

that is

$$b_1 : c_1 = b_1 : -(c_1), \text{ and this with } a_1 : b_1 : c_1 : d_1 \text{ gives all the ratios, or we have}$$

$$a_2 : b_2 : c_2 : d_2 = a_1(d_1 - c_1) : b_1(b_1 - c_1) : c_1(c_1 - d_1) : d_1(c_1 - a_1), \text{ \&c.}$$

We have then for example

$$b_2 - c_2 : c_2 - a_2 : a_2 - b_2 = b_1 - c_1 : c_1 - a_1 : a_1 - b_1; \text{ \&c.},$$

showing the identity of the forms in (a_1, b_1, c_1, d_1) and (a_2, b_2, c_2, d_2).

Article No. 194. Transformation to a New Set of Concyclic Foci.

194. Consider the equation

$$\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}} + \sqrt{p\mathfrak{D}} = 0,$$

which refers to the foci A, B, C, D; and taking the fourth concyclic focus, let (A_1, D_1) be the antipoints of (A, D) and (B_1, C_1) the antipoints of (B, C); so that (A_1, B_1, C_1, D_1) are another set of concyclic foci. Then the equation may be transformed so as to refer to the new set of foci; and the transformed equation is found to be

$$\sqrt{l_1\mathfrak{A}_1} + \sqrt{m_1\mathfrak{B}_1} + \sqrt{n_1\mathfrak{C}_1} + \sqrt{p_1\mathfrak{D}_1} = 0,$$

where l_1, m_1, n_1, p_1 are linear functions of l, m, n, p, with coefficients depending on the distances between the old and new foci.

195. I. The general case; l, m, n, p not subjected to any condition. The curve is here of the order $= 8$; it has a quadruple point at each of the points I, J (and there is consequently no other point at infinity); the number of nodes is $= 6$. Hence dps. $= 6 + 2 \cdot 10 = 26$; class is $= 12$; deficiency $= 3$.

196. II. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$; order of tetrazomal $= 6$; and hence order of each of the trizomals is $= 3$. There is here a single branch containing $(z^2 = 0)$ the line infinity twice; the order is $= 6$. Each of the points I, J is a double point, and there are therefore two more points at infinity, that is (besides the asymptotes at I, J), there are two (real or imaginary) asymptotes. The number of nodes, as in the general case, is $= 6$. Hence dps. $= 6 + 2 \cdot 1 = 8$; class is $= 14$; deficiency $= 2$. I notice the included particular case where the circles reduce themselves to their centres; viz., we have here the curve

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

which (see ante No. 93) is in fact the curve which is the locus of the foci of the conics which pass through the four points A, B, C, D. It is at present assumed that the four points are not in a circle; this case will be considered post No. 200.

197. III. $\sqrt{l}+\sqrt{m}=0$, $\sqrt{n}+\sqrt{p}=0$; order of the tetrazomal is $=6$; and hence order of each of the trizomals is $=3$. To verify this, observe that here

$$(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p}) = (\sqrt{l}, -\sqrt{l}, \sqrt{n}, -\sqrt{n}),$$

which since $a+b+c+d=0$ gives $l:m:n:p=a:b:c:d$; and the equation becomes

$$\sqrt{l}(\sqrt{\mathfrak{A}}-\sqrt{\mathfrak{B}})+\sqrt{n}(\sqrt{\mathfrak{C}}-\sqrt{\mathfrak{D}})=0,$$

which is a pair of trizomals, each of order 3.

198. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; corresponds to IV. supra, viz., there is a branch ideally containing $(z^2=0)$ the line infinity twice. But, observe that whereas in IV. supra, in order that this might be so, it was necessary to impose on l, m, n, p three conditions giving the definite systems of values $\sqrt{l}:\sqrt{m}:\sqrt{n}:\sqrt{p}=a:b:c:d$, in the present case only two conditions are imposed, so that a single arbitrary parameter is left.

199. V. $\sqrt{l}=\sqrt{m}=\sqrt{n}=\sqrt{p}$; corresponds to V. supra.

200. VI. $\sqrt{l}+\sqrt{m}=0$, $\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$, or what is the same thing, $\sqrt{l}:\sqrt{m}:\sqrt{n}:\sqrt{p}=c-d:d-c:b-a:a-b$; the equation is thus $(c-d)(\sqrt{\mathfrak{A}}-\sqrt{\mathfrak{B}})-(a-b)(\sqrt{\mathfrak{C}}-\sqrt{\mathfrak{D}})=0$. There is here one branch ideally containing $(z^2=0)$ the line infinity twice.

201. VII. If we have $l, m, n, p = 0$, and $(1, 1, 1, 1)(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p}) = 0$, see ante, Nos. 87 *et seq.* it there appears that (unless the centres A, B, C, are in a line) the condition signifies that the four circles have a common orthotomic circle; and when we have also

$$\frac{ab - cd}{l} + \frac{cd - ab}{m} + \frac{ac - bd}{n} + \frac{bd - ac}{p} = 0.$$

The formulæ for the decomposition are given ante, Nos. 42 *et seq.* Writing therein $\mathfrak{A}^\circ$, $\mathfrak{B}^\circ$, $\mathfrak{C}^\circ$, $\mathfrak{D}^\circ$ in place of $\mathfrak{U}$, $\mathfrak{V}$, $\mathfrak{W}$, $\mathfrak{T}$ respectively, it thereby appears that the tetra- zomal curve $\sqrt{l\mathfrak{A}^\circ} + \sqrt{m\mathfrak{B}^\circ} + \sqrt{n\mathfrak{C}^\circ} + \sqrt{p\mathfrak{D}^\circ} = 0$, breaks up into the two trizomal curves

$$\sqrt{\lambda\mathfrak{A}^\circ} + \sqrt{\mu\mathfrak{B}^\circ} + \sqrt{\nu\mathfrak{C}^\circ} = 0, \quad \sqrt{\lambda'\mathfrak{A}^\circ} + \sqrt{\mu'\mathfrak{B}^\circ} + \sqrt{\nu'\mathfrak{C}^\circ} = 0,$$

where

$$\lambda = l + m, \quad \mu = m + n, \quad \nu = n + l,$$

and where we have

$$\lambda' = -l + m, \quad \mu' = -m + n, \quad \nu' = -n + l.$$

Article Nos. 198 to 203. Cases of the Decomposable Curve, Centres not in a line.

198. I assume, in the first instance, that the centres of the circles are not in a line. The tetrazomal curve $\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}} + \sqrt{p\mathfrak{D}} = 0$ breaks up into two trizomals when the coefficients (l, m, n, p) satisfy certain conditions; and the forms of the trizomals are as follows.

199. The several cases of the decomposable curve are as follows:
I. The general case; l, m, n, p not subjected to any condition. The curve is here of the order $= 8$; it has a quadruple point at each of the points I, J (and there is consequently no other point at infinity); the number of nodes is $= 6$. Hence dps. $= 6 + 2 \cdot 10 = 26$; class is $= 12$; deficiency $= 3$.
II. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$; order of tetrazomal $= 6$; and hence order of each of the trizomals is $= 3$. There is here a single branch containing $(z^2 = 0)$ the line infinity twice; the order is $= 6$. Each of the points I, J is a double point, and there are therefore two more points at infinity. The number of nodes is $= 6$. Hence dps. $= 6 + 2 \cdot 1 = 8$; class is $= 14$; deficiency $= 2$.
III. $\sqrt{l} + \sqrt{m} = 0$, $\sqrt{n} + \sqrt{p} = 0$; order of the tetrazomal is $= 6$; and hence order of each of the trizomals is $= 3$.
IV. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$, $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$; corresponds to IV. supra, viz., there is a branch ideally containing $(z^2 = 0)$ the line infinity twice. But in the present case only two conditions are imposed, so that a single arbitrary parameter is left.
V. $\sqrt{l} = \sqrt{m} = \sqrt{n} = \sqrt{p}$; corresponds to V. supra.
VI. $\sqrt{l} + \sqrt{m} = 0$, $\sqrt{n} + \sqrt{p} = 0$, $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$; the equation is $(c - d)(\sqrt{\mathfrak{A}} - \sqrt{\mathfrak{B}}) - (a - b)(\sqrt{\mathfrak{C}} - \sqrt{\mathfrak{D}}) = 0$. There is here one branch ideally containing $(z^2 = 0)$ the line infinity twice.

202. VII. If we have l, m, n, p satisfying the two conditions, and $(1, 1, 1, 1)$ with $(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p})$, see ante, Nos. 87 *et seq.* it there appears that (unless the centres A, B, C, are in a line) the condition signifies that the four circles have a common orthotomic circle; and when we have also

$$\frac{ab-cd}{l} + \frac{cd-ab}{m} + \frac{ac-bd}{n} + \frac{bd-ac}{p} = 0.$$

The formulæ for the decomposition are given ante, Nos. 42 *et seq.*
203. Writing therein $\mathfrak{A}^\circ$, $\mathfrak{B}^\circ$, $\mathfrak{C}^\circ$, $\mathfrak{D}^\circ$ in place of $\mathfrak{U}$, $\mathfrak{V}$, $\mathfrak{W}$, $\mathfrak{T}$ respectively, it thereby appears that the tetrazomal curve $\sqrt{l\mathfrak{A}^\circ} + \sqrt{m\mathfrak{B}^\circ} + \sqrt{n\mathfrak{C}^\circ} + \sqrt{p\mathfrak{D}^\circ} = 0$, breaks up into the two trizomal curves

$$\sqrt{\lambda\mathfrak{A}^\circ} + \sqrt{\mu\mathfrak{B}^\circ} + \sqrt{\nu\mathfrak{C}^\circ} = 0, \quad \sqrt{\lambda'\mathfrak{A}^\circ} + \sqrt{\mu'\mathfrak{B}^\circ} + \sqrt{\nu'\mathfrak{C}^\circ} = 0,$$

where

$$\lambda = l + m, \quad \mu = m + n, \quad \nu = n + l,$$

and

$$\lambda' = -l + m, \quad \mu' = -m + n, \quad \nu' = -n + l.$$

Article Nos. 204 to 207. Further Cases of the Decomposable Curve.
204. IV. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$, $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$; order of tetrazomal is $= 4$; orders of trizomals $= 3$ each; corresponding to IV. supra.
205. V. $\sqrt{l} = \sqrt{m} = \sqrt{n} = \sqrt{p}$; order of tetrazomal is $= 5$; orders of trizomals $= 3$ each.
206. And hence one or other of the two functions

$$\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}}, \quad \sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}}$$

is $= 0$; that is, one of the trizomal curves is a cubic.
207. III. $\sqrt{l} + \sqrt{p} = 0$, $\sqrt{m} + \sqrt{n} = 0$; order of the tetrazomal is $= 6$; and hence order of each of the trizomals is $= 3$. To verify this, observe that here

$$(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p}) = (\sqrt{l}, \sqrt{m}, -\sqrt{m}, -\sqrt{l}),$$

which since $a + b + c + d = 0$ gives the required ratios.

208. In the general case where the points A, B, C, D are not on a circle, the tetrazomal curve is indecomposable; and the order is $= 8$ in the general case, $= 6$ in the special case where $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$, and so on as above.

209. The decomposable cases arise when there exists a linear relation between the square roots of the focal distances; and this is the condition that the four foci shall lie on a circle. When this condition is satisfied, the tetrazomal curve breaks up into two trizomal curves, each of which is a cubic.

210. The several cases may be tabulated as follows:

Case	**Order of Tetrazomal**	**Order of Trizomals**
I. General	8	4, 4
II. $\sum \sqrt{l} = 0$	6	3, 3
III. $\sqrt{l} + \sqrt{m} = 0$, $\sqrt{n} + \sqrt{p} = 0$	6	3, 3
IV. $\sum \sqrt{l} = 0$, $\sum al = 0$	4	3, 3
V. $\sqrt{l} = \sqrt{m} = \sqrt{n} = \sqrt{p}$	5	3, 3
VI. $\sqrt{l} + \sqrt{m} = 0$, $\sqrt{n} + \sqrt{p} = 0$, $\sum a\sqrt{l} = 0$	6	3, 3

211. The condition that the tetrazomal curve shall be decomposable is that the four foci shall lie on a circle. This condition may be expressed analytically as follows. Let the coordinates of the foci be (x_r, y_r), $r = 1, 2, 3, 4$; then the condition that they shall lie on a circle is

$$\begin{vmatrix} x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \\ x_4^2 + y_4^2 & x_4 & y_4 & 1 \end{vmatrix} = 0.$$

When this condition is satisfied, the tetrazomal curve breaks up into two trizomals.

212. The condition may also be expressed in terms of the distances between the foci. If d_{12}, d_{13}, &c. are the distances between the pairs of foci, then the condition is

$$d_{12}d_{34} + d_{13}d_{24} + d_{14}d_{23} = 0,$$

which is the condition that four points shall lie on a circle.

213. When the four foci lie on a circle, the tetrazomal curve is said to be "symmetrical;" and the two trizomals into which it breaks up are called the "components" of the tetrazomal. The components are each of them a circular cubic, and they have the same foci as the original tetrazomal.

214. The equation $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$ gives $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, and combining with $a+b+c+d=0$, we obtain the ratios of $l:m:n:p$ in terms of $a:b:c:d$. The resulting equation is

$$(b-c)\sqrt{\mathfrak{A}}+(c-a)\sqrt{\mathfrak{B}}+(a-b)\sqrt{\mathfrak{C}}=0,$$

which is the equation of a circular cubic. The tetrazomal curve thus breaks up into two trizomals, each of which is a circular cubic.

215. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; order of tetrazomal is $=4$; orders of trizomals $=3$ each; corresponding to IV. supra. The two conditions determine the ratios $l:m:n:p$ uniquely, and the tetrazomal is completely determined.

216. The several cases of the decomposable curve may be distinguished by the number of conditions imposed on the coefficients (l,m,n,p). In Case I. there are no conditions; in Case II. there is one condition; in Case III. there are two conditions; and in Case IV. there are three conditions. The greater the number of conditions, the lower the order of the tetrazomal.

217. The general theory of polyazomal curves includes the theory of the tetrazomal as a particular case. A polyazomal curve of order n is defined by the equation

$$\sum_{r=1}^{n} a_r \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r = 0$ is the equation of a pair of lines. The curve is of order $2n$, and it has n foci. The theory of the focal circles, the focal conic, and the bitangents may be generalized to these curves.

218. The condition that a polyazomal curve shall break up into curves of lower order is that the foci shall lie on a conic. When this condition is satisfied, the curve breaks up into two curves of order n, each having the same foci as the original curve.

219. The number of conditions required for the decomposition of a polyazomal curve is $n - 1$. Thus for a tetrazomal ($n = 4$) the number of conditions is 3; for a pentazomal ($n = 5$) it is 4; and so on. The conditions are linear in the coefficients $(a_1, a_2, \ldots, a_n)$.

220. The general theory of polyazomal curves is thus seen to be a natural generalization of the theory of the bicircular quartic. The methods employed in the investigation of the bicircular quartic may be applied, with suitable modifications, to the investigation of polyazomal curves of any order.

221. The memoir concludes with a summary of the principal results obtained. The theory of polyazomal curves has been developed from the fundamental conception of the focus of an algebraic curve; and it has been shown that the focal properties of these curves are intimately connected with the theory of transformations, and with the general theory of algebraic curves.

222. The principal results are as follows:

(1) Every algebraic curve of order n has n^2 foci, which are the intersections of the tangents from the circular points at infinity.

(2) The equation of the curve may be written in the form $\sum \sqrt{\mathfrak{A}_r} = 0$, where $\mathfrak{A}_r = 0$ is the equation of the pair of tangents from the rth focus.

(3) The focal circles of the curve are circles passing through $n-1$ of the n^2 foci; and they have a common orthogonal circle.

(4) The condition that the curve shall have a node or a cusp is that two or three of the foci shall coincide.

(5) The condition that the curve shall break up into curves of lower order is that the foci shall lie on a conic.

223. The theory of polyazomal curves is capable of further extension; and it is hoped that the present investigation may serve as a foundation for future researches on this interesting subject.

[The memoir continues with additional notes and references to previous work.]

224. And then

$$(d-a)\sqrt{l} = (b-d)\sqrt{m} + (c-d)\sqrt{n},$$
$$(d-a)\sqrt{p} = (a-b)\sqrt{m} + (a-c)\sqrt{n},$$

and substituting these values in the equation $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$, we obtain the condition for the decomposition.

225. $\sqrt{l} = \sqrt{m} = \sqrt{n} = \sqrt{p}$; order of tetrazomal is $= 5$; orders of trizomals $= 3$ each. In this case the four foci are concyclic, and the tetrazomal breaks up into two circular cubics.

Article Nos. 226 to 230. Cases of the Decomposable Curve, Centres in a line.

226. I proceed to consider the case where the centres of the circles are in a line. In this case the tetrazomal curve is necessarily decomposable; and the components are of lower order than in the general case.

227. If the centres A, B, C, D are in a line, then the four circles have a common orthogonal circle only if the line of centres passes through the centre of the orthogonal circle. In this case the tetrazomal breaks up into two trizomals, each of order 3.

228. If the line of centres does not pass through the centre of the orthogonal circle, then the tetrazomal is indecomposable; but it may have a node or a cusp at one of the points at infinity.

229. The case where the centres are in a line is of importance for the theory of the symmetrical bicircular quartic. In this case the focal axis is the line of centres; and the four foci lie on this line. The curve has an axis of symmetry, and the focal circles are symmetrically situated with respect to this axis.

230. The investigation of this case leads to the consideration of the Cartesian curve as a limiting case of the bicircular quartic. When the four foci are in a line, the bicircular quartic degenerates into a Cartesian; and the focal properties undergo the degradation already referred to.

231. The case where the centres are in a line may be further subdivided according to the relative positions of the centres. If the centres are in order A, B, C, D on the line, then the distances satisfy certain inequalities; and the nature of the curve depends on these inequalities.

232. If the centres are such that $AB \cdot CD = AC \cdot BD = AD \cdot BC$, then the four points form a harmonic range; and the tetrazomal curve has special properties. In this case the curve is symmetrical, and the focal circles are equal.

233. The condition that four points on a line shall form a harmonic range may be expressed in various ways. If the distances from a fixed point on the line are a, b, c, d, then the condition is

$$\frac{2}{b-c} = \frac{1}{a-b} + \frac{1}{d-c},$$

or, what is the same thing,

$$(a+d)(b+c) = 2(ad+bc).$$

When this condition is satisfied, the tetrazomal curve has a node at each of the circular points at infinity.

234. The case of harmonic centres is of interest in connexion with the theory of the bicircular quartic. The harmonic relation between the foci leads to simplifications in the equation of the curve; and the focal circles have properties analogous to those of the circles of Apollonius.

235. The memoir concludes with the indication of certain directions in which the theory may be extended. The theory of polyazomal curves is capable of generalization in many ways; and it is hoped that the present investigation may serve as a foundation for future researches on this interesting subject.

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236. *Note on the Focal Properties of the Bicircular Quartic.*—The foregoing theory of the focal properties of the bicircular quartic may be presented under a somewhat different form, which brings into clearer view the connexion with the theory of the conic.

237. If we regard the bicircular quartic as the envelope of a circle which cuts three given circles at given angles, then the centre of the variable circle describes a conic; and the foci of the bicircular quartic are the centres of the three given circles. This is the fundamental theorem of the focal theory.

238. The theorem may be proved analytically as follows. Let the three given circles be

$$x^2 + y^2 + 2g_r x + 2f_r y + c_r = 0, \quad (r = 1, 2, 3),$$

and let the variable circle be

$$(x - \xi)^2 + (y - \eta)^2 = \rho^2.$$

The condition that the variable circle shall cut the rth given circle at a given angle θ_r is

$$\rho^2 + r_r^2 - d_r^2 = 2\rho r_r \cos\theta_r,$$

where r_r is the radius of the rth circle, and d_r is the distance between the centres. Eliminating ρ between these three equations, we obtain the locus of (ξ, η), which is a conic.

239. The conic which is the locus of the centre of the variable circle is the "focal conic" of the system of circles. It passes through the centres of the three given circles; and its properties are analogous to those of the director circle of a conic.

240. The focal conic may be used to investigate the properties of the bicircular quartic. If four circles have a common orthogonal circle, then the four centres lie on a conic; and conversely, if four points lie on a conic, then the four circles having these points as centres and cutting a given circle orthogonally have a common orthogonal circle.

241. The theorem of the focal conic leads to a simple proof of Casey's equation for four circles touching a fifth. If four circles touch a fifth circle, then their centres lie on a conic; and the condition that four points shall lie on a conic is equivalent to Casey's equation.

242. The focal properties of the bicircular quartic are thus seen to be intimately connected with the theory of conics and circles. The theory provides a unified point of view from which many apparently disconnected theorems may be regarded as corollaries of a single fundamental principle.

Article Nos. 243 to 245. Applications to the Theory of Curves.

243. The theory of polyazomal curves has many applications to the theory of algebraic curves. In particular, it furnishes a method for the investigation of curves with given singularities, and for the classification of curves of a given order.

244. The method consists in regarding the curve as the locus of a point whose distances from fixed points satisfy a given relation. The fixed points are the foci of the curve; and the given relation is the focal equation. By varying the relation, we obtain a system of curves with the same foci but different singularities.

245. The theory may also be applied to the investigation of the bitangents and inflexions of a curve. The bitangents correspond to the double tangents from the foci; and the inflexions correspond to the points where the focal circles touch the curve.

246. The classification of curves by means of their focal properties is as follows. A curve of order n has n^2 foci; and the arrangement of these foci determines the nature of the curve. If the foci are distinct, the curve is general; if some of the foci coincide, the curve has singularities; and if the foci lie on a conic, the curve is degenerate.

247. The number of foci which must coincide in order that the curve shall have a node or a cusp may be determined as follows. For a node, two foci must coincide; for a cusp, three foci must coincide; and for a tacnode, four foci must coincide. The greater the number of coincident foci, the higher the singularity.

248. The condition that a curve shall have a given number of nodes and cusps may be expressed as a system of equations between the focal coordinates. These equations are algebraic; and their solution gives the focal conditions for the given singularities.

249. The theory of focal singularities is thus seen to be a powerful instrument for the investigation of algebraic curves. It provides a method for the systematic discussion of curves with given singularities; and it leads to results which are not easily obtainable by other methods.

Article Nos. 250 to 252. Conclusion.

250. The present memoir has developed the theory of polyazomal curves from the fundamental conception of the focus of an algebraic curve. It has been shown that the focal properties of these curves are intimately connected with the theory of transformations, and with the general theory of algebraic curves.

251. The principal results of the investigation are: the theorem that every algebraic curve of order n has n^2 foci; the theorem that the equation of the curve may be written in the form $\sum \sqrt{\mathfrak{A}_r} = 0$; the theorem of the focal circles; and the theorem of the decomposition of polyazomal curves when the foci lie on a conic.

252. The theory is capable of further extension; and it is hoped that it may serve as a foundation for future researches on the focal properties of algebraic curves.

253. The condition that the four circles shall have a common orthogonal circle is that the distances of their centres from any line shall satisfy a certain linear relation. This condition may be written in the form

$$\begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix} = 0,$$

where a, b, c are the distances of the three centres from any line. The condition is equivalent to the condition that the three centres shall lie on a conic.

254. When the four circles have a common orthogonal circle, the tetrazomal curve breaks up into two trizomals; and each of these is a circular cubic. The four foci of the tetrazomal are the centres of the four circles; and the three foci of each trizomal are three of these four centres.

255. The condition for the decomposition may also be expressed as follows. If the four circles are A, B, C, D, and if S is their common orthogonal circle, then the radical axis of any two of the circles passes through the centre of S. The four radical axes thus meet in a point, which is the centre of the common orthogonal circle.

256. The theorem that four circles have a common orthogonal circle if and only if their centres lie on a conic is a fundamental theorem of the theory. It is the generalization of the theorem that three circles have a common orthogonal circle if and only if their centres are collinear.

257. The theorem may be proved analytically as follows. Let the equations of the four circles be

$$x^2 + y^2 + 2g_r x + 2f_r y + c_r = 0, \quad (r = 1, 2, 3, 4).$$

The condition that these shall have a common orthogonal circle is that there shall exist values of G, F, C such that

$$(G - g_r)^2 + (F - f_r)^2 = C - c_r, \quad (r = 1, 2, 3, 4).$$

Eliminating G, F, C, we obtain the condition that the four circles shall have a common orthogonal circle.

258. The condition thus obtained is equivalent to the condition that the four points (g_r, f_r) shall lie on a conic. For the condition that four points shall lie on a conic is

$$\begin{vmatrix} g_1^2+f_1^2 & g_1 & f_1 & 1 \\ g_2^2+f_2^2 & g_2 & f_2 & 1 \\ g_3^2+f_3^2 & g_3 & f_3 & 1 \\ g_4^2+f_4^2 & g_4 & f_4 & 1 \end{vmatrix} = 0;$$

and this is precisely the condition that the four circles shall have a common orthogonal circle.
259. The theorem is thus established; and it forms the foundation of the theory of the focal properties of polyazomal curves. The theorem that the tetrazomal curve breaks up into two trizomals when the four foci lie on a circle is a particular case of this general theorem.
260. The investigation of the present memoir has been confined to the consideration of curves of the fourth order; but the methods employed are applicable to curves of any order. The theory of polyazomal curves of order n presents many interesting problems, which it is hoped to investigate in a future memoir.

Article Nos. 261 to 263. Note on the Focal Conic.

261. The focal conic of a system of circles has been defined as the locus of the centre of a circle which cuts the given circles at given angles. The focal conic passes through the centres of the given circles; and its properties are analogous to those of the director circle of a conic.
262. The equation of the focal conic may be obtained as follows. Let the given circles be

$$x^2+y^2+2g_rx+2f_ry+c_r=0, \quad (r=1,2,3).$$

The condition that the variable circle $(x-\xi)^2+(y-\eta)^2=\rho^2$ shall cut the rth circle at an angle θ_r is

$$\rho^2+r_r^2-d_r^2=2\rho r_r\cos\theta_r.$$

Eliminating ρ between these three equations, we obtain the locus of (ξ,η), which is the focal conic.
263. The focal conic is thus seen to be a conic passing through the centres of the three given circles. Its equation is of the second degree in (ξ,η); and its coefficients depend on the radii and the angles of intersection.

264. The focal conic has many interesting properties. In particular, it is the locus of the centre of a circle which cuts three given circles orthogonally; and it is the envelope of the radical axis of two circles which cut the three given circles at given angles.
265. The focal conic is also the locus of a point whose distances from the centres of the three given circles are proportional to the radii of those circles. This property is analogous to the property of the director circle of a conic, which is the locus of a point whose distances from the foci are equal.
266. The focal conic may be used to investigate the properties of the system of circles. If four circles have a common orthogonal circle, then the focal conics of any three of them pass through the centre of the fourth; and conversely, if the focal conics of three circles pass through a point, then the four circles have a common orthogonal circle.
267. The theory of the focal conic is thus seen to be an important part of the theory of circles and their orthogonal trajectories. It provides a method for the investigation of the properties of systems of circles; and it leads to many elegant theorems in geometry.

Article Nos. 268 to 270. Applications to the Problem of Apollonius.

268. The problem of Apollonius,—to describe a circle touching three given circles,—is one of the most famous problems in geometry. The solution by means of the focal conic is as follows.
269. Let the three given circles be A, B, C. The focal conic of these three circles passes through their centres; and the centre of the required circle is a point on the focal conic. The radius of the required circle is determined by the condition that it shall touch the three given circles.
270. The condition of tangency gives a quadratic equation for the radius; and the two values of the radius correspond to the two solutions of the problem. The eight solutions of the problem of Apollonius correspond to the eight points in which the focal conic is met by the three circles which are the loci of the centres of circles cutting the given circles at the supplementary angles.

271. The theorem that the eight solutions of the problem of Apollonius correspond to the eight foci of a bicircular quartic is a beautiful example of the connexion between the theory of circles and the theory of curves. The bicircular quartic which has the three given circles for its focal circles has the required touching circles for its bitangents; and the eight foci of the quartic are the centres of the eight touching circles.

272. The investigation of the problem of Apollonius by means of the focal conic leads to many generalizations. The problem may be extended to the case of four or more circles; and the solution may be obtained by means of the focal conic of the system.

273. The theory of the focal conic thus provides a unified method for the investigation of problems relating to circles and their contacts. It is a powerful instrument for the solution of geometrical problems; and it leads to results which are not easily obtainable by other methods.

Article Nos. 274 to 276. Generalization of the Focal Theory.

274. The theory of foci may be generalized as follows. Instead of considering the intersections of the tangents from the circular points at infinity, we may consider the intersections of the tangents from any two fixed points. These intersections may be called the "generalized foci" of the curve; and the theory of these foci is analogous to the theory of the ordinary foci.

275. If the two fixed points are P and Q, then the generalized foci of a curve of order n are the intersections of the tangents from P and Q to the curve. The number of generalized foci is n^2; and their arrangement determines the nature of the curve.

276. The theory of generalized foci may be applied to the investigation of curves with given properties. In particular, it leads to a generalization of the theorem that the sum of the distances from any point of a conic to the two foci is constant. The generalized theorem is that the sum of the distances from any point of a curve to the two fixed points P and Q, measured along the tangents to the curve, is constant.

277. The condition that four circles shall have a common orthogonal circle may be expressed in terms of the distances of their centres. If the centres are A, B, C, D, and the distances are $AB = c$, $AC = b$, $AD = d$, $BC = a$, $BD = f$, $CD = e$, then the condition is

$$\begin{vmatrix} 0 & c^2 & b^2 & d^2 & 1 \\ c^2 & 0 & a^2 & f^2 & 1 \\ b^2 & a^2 & 0 & e^2 & 1 \\ d^2 & f^2 & e^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0.$$

This is the condition that the four points shall lie on a conic; and it is equivalent to the condition that the four circles shall have a common orthogonal circle.

278. The condition may also be written in the form

$$(ab + cd)(ac + bd)(ad + bc) = 0,$$

which shows that the condition is satisfied if $ab + cd = 0$, or $ac + bd = 0$, or $ad + bc = 0$. These are the conditions that the four points shall form a harmonic range.

279. The general theory of polyazomal curves includes the theory of the bicircular quartic, the circular cubic, and the conic as particular cases. The methods of the theory are applicable to curves of any order; and they lead to results which are of great importance for the theory of algebraic curves.

280. The memoir concludes with the hope that the theory of polyazomal curves may serve as a foundation for future researches on the focal properties of algebraic curves. The theory is capable of extension in many directions; and it is believed that it will lead to important discoveries in the theory of curves and their applications.

End of the Memoir on Polyazomal Curves.